



ENGINEERING INTERNSHIP REPORT

CLOUD DEPLOYMENT ON VMware vSphere

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Chapter 1

General Introduction

1.1 Internship Objectives

This internship was an opportunity to apply academic knowledge to a practical engineering environment. Its main objective was to develop practical skills mainly in private cloud infrastructure deployment. To realize the objective, a layered deployment of the private cloud infrastructure was executed: physical layer, underlay fabric, overlay fabric, and the application layer, focusing on virtualization, networking and integrating VMware capabilities with Kubernetes. The main tasks included deploying a VMware vSphere environment, connecting the hardware and the virtual machines, preparing a Kubernetes environment, deploying the application and finally testing.

The internship also had a professional development objective. In addition to technical implementation, I learned how requirements are collected, how technical architecture is designed, how a project is managed, and how technical decisions and problems are documented.

1.2 Technical Terminology

- **Virtualization:** A technology that allows physical computing resources to be divided into virtual machines.
- **ESXi:** VMware's type-1 hypervisor used to run virtual machines directly on physical hardware.
- **VMware vSphere:** A virtualization platform used to manage ESXi hosts, virtual machines, clusters, storage, networking, and availability features.
- **vCenter Server:** The central management platform used to manage ESXi hosts and vSphere resources.
- **Kubernetes:** An open-source platform used to deploy, manage, and scale containerized applications.
- **Pod:** The basic execution unit in Kubernetes. A pod contains one or more containers.
- **PersistentVolume (PV):** A Kubernetes storage resource representing persistent storage available to workloads.

- **PersistentVolumeClaim (PVC):** A request made by a Kubernetes workload for persistent storage.
- **StorageClass:** A Kubernetes resource that defines how storage can be dynamically provisioned.
- **CSI:** Container Storage Interface. It provides a standard interface between Kubernetes and storage systems.
- **vSphere CSI Driver:** The VMware CSI implementation that allows Kubernetes to request and manage storage backed by VMware infrastructure.
- **Service:** A Kubernetes object that provides a stable way to access a group of pods.
- **Ingress:** A Kubernetes resource used to define HTTP or HTTPS access to services inside the cluster.
- **Spine-leaf architecture:** A data-center network architecture where leaf switches connect the subnets behind them to other networks through spine switches, providing predictable and scalable connectivity.

Chapter 2

Presentation of the Host Company

2.1 General Presentation of Huawei

Huawei was founded in 1987 and is a global provider of information and communications technology infrastructure and smart devices. The company operates in more than 170 countries and regions and has a large international workforce. Huawei develops technologies and solutions for telecommunications operators, enterprises, and consumers. Its activities include telecommunications networks, enterprise IT, cloud technologies, digital solutions, smart devices, and other technology-related fields.

2.2 Huawei Technologies Tunisia

Huawei Technologies Tunisia is part of Huawei's international organization and operates in the Tunisian technology and telecommunications environment. The Tunisian branch provides an environment where technical teams work on telecommunications, ICT infrastructure, cloud-related technologies, and enterprise solutions. The internship therefore took place in a professional environment where infrastructure and networking technologies are directly relevant to business activities. The internship also provided exposure to different engineering roles. I interacted with technical teams and project-management activities, which helped me understand that infrastructure projects require cooperation between several areas of expertise.

2.3 Working Environment

The internship started with meetings involving HR, my supervisor, and engineering teams. These meetings helped me understand the working environment, internal procedures, and expectations of the internship. The environment was culturally diverse, with people from different nationalities and professional backgrounds. This was useful for developing communication skills and understanding how technical teams cooperate in an international company. The first days were also used to prepare my workstation and become familiar with the company's premises and working methods. This provided the necessary foundation before beginning the technical project.

Chapter 3

Project Context

3.1 Problematic

The information and communication technology industry is increasingly based on virtualization, cloud computing, containerization, automation, and software-defined infrastructure. Organizations especially ones that are under strict regulations, like one of Huawei's clients subject to this report, need secure, highly available private infrastructure that can run different applications reliably without high costs. So, it is justifiable for one of Huawei's clients to request the implementation of such technologies on their premises.

3.2 Solution

Deploying a private on-premises datacenter to meet the previously stated goals requires the necessary hardware, and the software solution to virtualize and unify the resources of the hardware into a shared resource pool required for cloud usage. And for that exact need, Huawei offers Huawei Cloud Stack, a collection of services developed by Huawei on top of OpenStack technologies that together build private datacenter architectures for various needs across different industries, prioritizing security availability and scalability... This solution is a perfect fit for the Huawei's client, but since 2nd year interns' involvement in client projects is strictly against policies, both client's and Huawei's, as well as to the Huawei cloud stack solution itself, I tried to work on the same problem to further understand real clients' business needs, how to turn them into achievable solutions and finally implement and ensure their effectiveness, but on a different industry-leading solution: VMware vSphere, which executes the same business goal cleanly but with some variations in some technical and architectural aspects.

3.3 Scope of the Project

The internship covered the project management for the two month-period following the waterfall methodology as well as the technical implementation of the infrastructure and some aspects of workload-management layers with a lightweight application rather than the complete deployment of a production application. The infrastructure layer included VMware ESXi, vCenter Server, virtual networking, datastores, and vSphere cluster features. The workload layer included Kubernetes and the storage, networking, and mechanisms required for application deployment.

3.4 Project Methodology

3.4.1 Importance of Project Management

At the beginning of the internship, the main focus was naturally placed on the technical aspects of the project. But when I immediately tried to start with the technical implementation, I quickly realized that the that I don't have enough understanding of the ultimate goal of what I am implementing, because I also realized later that a technical project starts with a business objective rather than with a specific technology. Furthermore, prior discussions with experienced engineers across industries and backgrounds showed me that engineering work is not limited to executing predefined technical tasks. A successful engineer is also expected to understand requirements, evaluate alternatives, make decisions, manage constraints, anticipate risks, and communicate the state of a project through technical and non-technical documentations to different stakeholders. This observation led me to add project management as an important dimension of the internship. The objective was to understand how technical tasks are organized within a real engineering project and how to prevent the project from reaching the end of the internship in an unfinished state. To develop this understanding, I sought guidance from my supervisor, Mr. Mohamed Ali Boujelben, a telecom project manager at Huawei and studied the different stages involved in transforming a high-level business need into an executable technical project. Although his availability was limited by their professional responsibilities, the discussions and observations provided useful insight into the relationship between technical engineering and project management. One of the first lessons was how to understand the client's business needs and requirements, turn them into technical requirements and conceptualize the adequate solution, the implementation of which will be following a PM methodology which would be determined on multiple factors.

3.4.2 From Business Objectives to Technical Requirements

A meeting with the client ends with this high level requirement statement: "We need to deploy a mission-critical application that processes sensitive, regulated data" A requirement such as "mission-critical application" must be evaluated and interpreted into a lower level requirement such as "the application must always be available" which is still not directly implementable by an engineer. It must also be transformed into measurable technical requirements. This process involves requirements elicitation, documentation, decomposition, feasibility analysis, and formal validation, which was done through the first half of the internship for the whole project.

Requirements

The requirements can generally be divided into three categories.

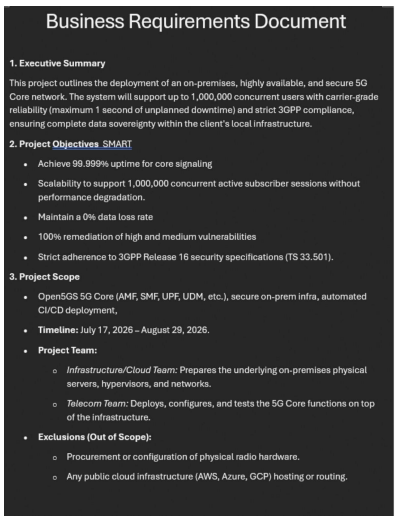
- **Functional requirements** describe what the system must do. In the context of this project, they included providing virtualized infrastructure, running Kubernetes workloads, provisioning persistent storage, exposing applications through the network, and supporting workload recovery and distribution.
- **Non-functional requirements** describe how the system should operate. These included availability, security, scalability, maintainability, controlled deployment, monitoring, and validation.

- **Constraints** describe the conditions under which the project must be completed. In this internship, important constraints included the available hardware, limited computing resources, the internship duration, the absence of a production customer environment, the inability to use public Cloud Service Providers (CSPs) and the need to reproduce enterprise concepts in a laboratory environment.

This decomposition bridges the gap between business expectations and engineering decisions ensuring alignment between parties in a professional customer project.

Documentation

I documented and validated these requirements through documents such as a Business Requirements Document (BRD) and a Software Requirements Specification (SRS)... Producing these documents was compliant to IEEE 1016 technical documentation criteria.



Business Requirements Document

1. Executive Summary

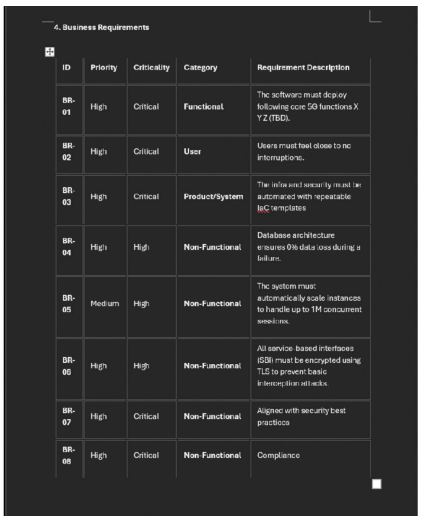
This project outlines the deployment of an on-premises, highly available, and secure 5G Core network. The system will support up to 1,000,000 concurrent users with carrier-grade reliability (maximum 1 second of unplanned downtime) and strict 3GPP compliance, ensuring complete data sovereignty within the client's local infrastructure.

2. Project Objectives SMART

- Achieve 99.999% uptime for core signaling
- Scalability to support 1,000,000 concurrent active subscriber sessions without performance degradation.
- Maintain a 0% data loss rate
- 100% remediation of high and medium vulnerabilities
- Strict adherence to 3GPP Release 16 security specifications (TS 33.501).

3. Project Scope

- Open5GS 5G Core (AMF, SMF, UPF, UDM, etc.), secure on-prem infra, automated CI/CD deployment,
- **Timeline:** July 17, 2026 – August 29, 2026.
- **Project Team:**
  - *Infrastructure/Cloud Team:* Prepares the underlying on-premises physical servers, hypervisors, and networks.
  - *Telecom Team:* Deploys, configures, and tests the 5G Core functions on top of the infrastructure.
- **Exclusions (Out of Scope):**
  - Procurement or configuration of physical radio hardware.
  - Any public cloud infrastructure (AWS, Azure, GCP) hosting or routing.



4. Business Requirements

ID	Priority	Criticality	Category	Requirement Description
BR-01	High	Critical	Functional	The software must deploy following core 5G functions X Y Z (TBD).
BR-02	High	Critical	User	Users must feel close to no interruptions.
BR-03	High	Critical	Product/System	The infra and security must be automated with repeatable IaC templates
BR-04	High	High	Non-Functional	Database architecture ensures 0% data loss during a failure.
BR-05	Medium	High	Non-Functional	The system must automatically scale instances to handle up to 1M concurrent sessions.
BR-06	High	High	Non-Functional	All service-based interfaces (SBI) must be encrypted using TLS to prevent basic interception attacks.
BR-07	High	Critical	Non-Functional	Aligned with security best practices
BR-08	High	Critical	Non-Functional	Compliance

Figure 3.1: Excerpt from Business Requirements Document 1.0 (Uploaded on GitHub for more details: https://github.com/4min4yed/Project-Management)

The BRD was used to express the project objectives and expected capabilities at a higher level, while the SRS translated these objectives into more precise technical and functional requirements. These documents served as references for determining what had to be implemented and tested to avoid conflicts between teams during implementation and they serve as the source of truth for all phases of the project implementation. The requirements phase therefore established a clear target before significant implementation work began. One of the main documents was the Software Design Document, which delves into the technical details of the project and will be talked about further down the report, was the most challenging to produce requiring the most technical knowledge to know what is adequate for the needs and requirements and what is not. So, most of the technical research was done alongside producing this document. Which quickly taught me that the SDD and other documents are not final and they can always be revisited and edited to further match more reuiqrements or workaround more constraints and problems that were not clear from the beginning or became visible further down the line of the project timeline. This point may be one of the main differentiators between good and better project managers. it is their ability to anticipate changes and problems as much as they could both in the technical and human aspects of the project, thus guaranteeing efficiency, result deliverability and client satisfaction.

3.4.3 Project Scope and Feasibility

Another important project-management concept studied during the internship was scope management. Before implementation, it is necessary to determine what is included in the project and what is outside its boundaries. This is particularly important for a short project timeframe like in the case of this internship because an ambitious technical scope can easily become impossible to complete within the available time, which was the case for my project, as unfortunately the project was not properly studied by my technical supervisor considering my allocated timeframe. The project scope was therefore limited to the infrastructure and some Kubernetes capabilities necessary to demonstrate the main objectives: VMware virtualization and Kubernetes deployment. Additional capabilities such as complete production monitoring, enterprise-scale automation, disaster recovery, and large-scale performance benchmarking were considered as future extensions rather than requirements for the core proof of concept. This distinction was important because it prevented secondary features set by the supervisor from consuming time that was required for the core implementation and validation.

3.4.4 Selection of the Project Methodology

Several project-management approaches were studied, including Agile, Scrum, Kanban, Extreme Programming, and traditional sequential methodologies to better understand which methodology best fits my project. Agile approaches are particularly useful when requirements evolve frequently and when continuous customer feedback is central to the project. Scrum organizes work into fixed-length sprints and introduces roles such as Product Owner and Scrum Master. Kanban emphasizes visual management and work-in-progress limits. While Extreme Programming gives particular importance to continuous testing, and frequent customer involvement. A traditional sequential methodology, commonly represented by the Waterfall model, follows a more structured progression from requirements analysis to design, implementation, testing, and deployment. For this internship, I selected a Waterfall-inspired methodology. The choice was mainly motivated by the nature of the project. The project had a clearly defined technical objective from the start, a limited duration of approximately eight weeks, and no external customer continuously changing the requirements during implementation. In addition, several technical components had strong dependencies: for example the Kubernetes cluster itself could not be meaningfully deployed before the VMware infrastructure and network were functional, which demands a sequential methodlogy for best results. However, the methodology was not interpreted as an absolute rule preventing modifications. Practical implementation inevitably revealed problems and constraints that required returning to earlier decisions. The project therefore followed a Waterfall structure while allowing controlled corrections when experience showed that an initial assumption or configuration was incorrect.

3.4.5 Project Planning and Gantt Dashboard

Once the methodology had been selected, the project was divided into phases and milestones. The main phases were research & requirements, architecture design, infrastructure deployment, and the Kubernetes integration. An interactive Gantt dashboard was developed using html & css to visualize the planned activities over the internship period, which made it possible to track the expected duration of tasks, their dependencies, and the progression of the project, thus reducing the risk of task creep.

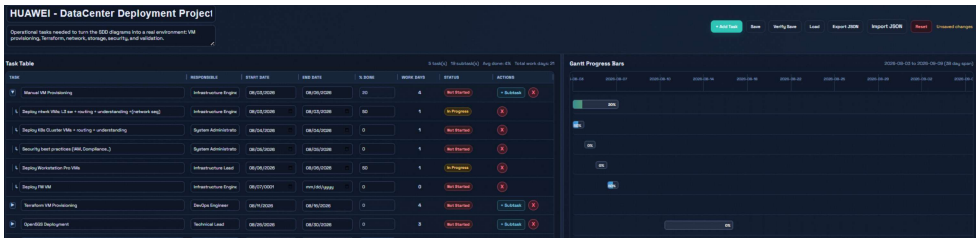


Figure 3.2: Screenshot of developed Gantt-Chart UI (Source code uploaded on GitHub for more details: https://github.com/4min4yed/Project-Management)

The use of the dashboard also helped identify a practical problem that became increasingly important during implementation: the difference between estimated task duration and actual task duration. At the beginning of the project, several activities were estimated relatively optimistically because I had not yet performed them in practice. Once implementation started, some tasks took considerably longer than expected because of configuration errors, compatibility problems, troubleshooting, and the need to understand unfamiliar technologies. This was particularly noticeable during the VMware infrastructure deployment. Problems involving hardware compatibility, nested virtualization, vCenter deployment, and virtual networking required several attempts before reaching a stable configuration. Consequently, a significant amount of time was spent not on the final configuration itself, but on identifying and correcting mistakes made during the first attempts. This was an important practical lesson: planning must account not only for the ideal implementation procedure but also for learning time, troubleshooting, reconfiguration, and validation. The Gantt dashboard was therefore used as a planning and monitoring tool rather than as a rigid schedule. When an activity took longer than expected, subsequent tasks were reviewed and priorities were adjusted to preserve the completion of the core project objectives.

3.4.6 Requirements Phase

Following the selected methodology, the first formal project phase was requirements analysis. The project requirements were organized into functional requirements, non-functional requirements, and constraints. The functional requirements focused on the capabilities that the infrastructure had to provide. The platform had to support virtualization, Kubernetes workloads, persistent storage, network exposure, ingress routing, and high-availability mechanisms. The non-functional requirements addressed the expected characteristics of the solution. The infrastructure had to be reliable, sufficiently secure, maintainable, and capable of supporting workload recovery and distribution. The deployment and validation process also had to be documented. The requirements were documented in the BRD and SRS prepared during this phase. The BRD established the project's objectives and overall expected outcome, while the SRS provided the technical interpretation of these objectives. The requirements documents consequently became the reference point for the next phase: system design.

3.4.7 Design Phase

After defining what the system had to provide, the next question was how these requirements would be implemented technically. This phase demands technical expertise in the

field of project, which in my case a deep technical research was crucial to make the correct design decisions, the research outcomes will be discussed later in this document. The design phase resulted in the preparation of a Software Design Document (SDD). The SDD was intended to provide a technical reference for the architecture and implementation of the system. The document was structured around both high-level and detailed low-level design considerations. At the high level, it described the overall architecture and the relationship between the physical infrastructure, virtualization platform, network, Kubernetes cluster, and application workloads. At the lower level, it described the configuration principles and responsibilities of the individual components. The design considered several dimensions: The physical architecture described the infrastructure available for the project and the way physical resources were expected to support the virtual environment. The logical architecture described the virtualization and workload layers built above the physical resources. The network architecture described the communication between infrastructure components. The security architecture considered access control and secure communication as cross-cutting concerns rather than isolated components. The SDD therefore served as the bridge between the requirements documents and the actual implementation. It defined the architecture that would be implemented, tested, and later evaluated.

3.4.8 Iterative Correction During Implementation

Although the project followed a sequential methodology, the practical work demonstrated that engineering is not perfectly linear. The first implementation attempts often exposed information that was not available during the design phase. A configuration that appears correct theoretically may behave differently because of hardware, software versions, networking, resource limitations, or dependencies between components. Several examples occurred during the VMware deployment. The initial ESXi installation encountered a network-driver compatibility problem. The vCenter deployment required several attempts because of the limitations of the laboratory environment. Nested virtualization introduced additional CPU and networking constraints. The virtual network also required adjustments before the expected communication behavior was obtained. These problems resulted in some rework and consumed significantly more time than initially planned. This experience demonstrated that the purpose of a design document is not to provide a baseline from which changes can be understood and controlled. When a contradiction is found durint implementation with the SDD, the corresponding design was reviewed rather than forcing or continuing with an invalid configuration. The project therefore maintained a sequential overall structure while allowing feedback from implementation and testing to improve the design.

Chapter 4

Research Phase

4.1 Cloud Computing and Private Cloud

Cloud computing can be understood as the abstraction of physical infrastructure into shared, manageable, and on-demand resources. Instead of assigning one physical server to one application permanently, virtualization allows several virtual machines to share the physical resources of a host. A cloud management platform can then automate provisioning, networking, storage allocation, identity, monitoring, and lifecycle operations. Huawei's own cloud documentation describes the cloud architecture as a set of layers including physical infrastructure, platform capabilities, applications, and management/O&M functions. For private-cloud deployments, this model is particularly useful because organizations can preserve control over physical infrastructure while adopting cloud-style resource management.

4.2 Huawei Cloud Stack

Huawei Cloud Stack provides a private and hybrid cloud environment built from multiple services. During the internship, I studied HCS not as a collection of isolated services, but as a layered platform:

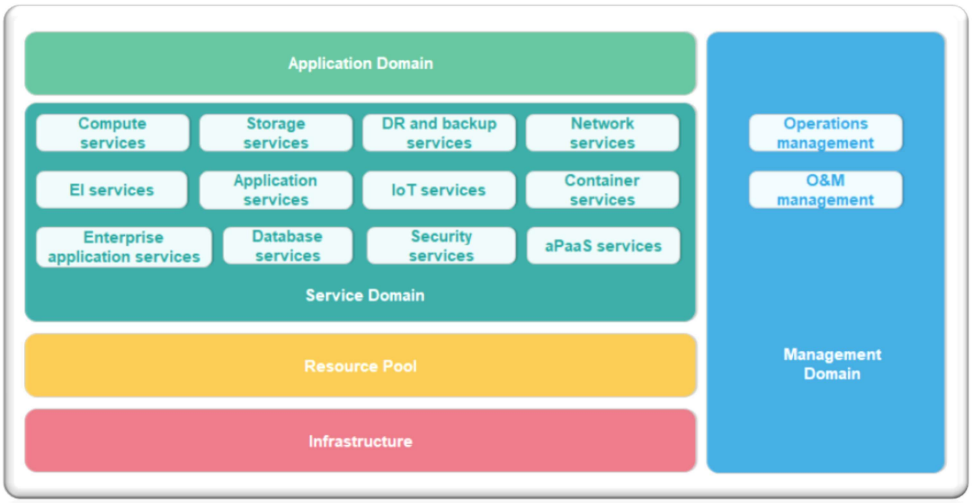


Figure 4.1: HCS Layered Architecture Diagram

At the infrastructure level are servers, switches, and storage arrays. Above this layer are compute, storage, and networking services. The virtualization and cloud-management layer then abstracts these physical resources into a shared pool. Above it are cloud services such as virtual machines, storage services, networking, containers, databases, security functions, and application services. This layered approach was one of the most important concepts I learned because it explains how a cloud platform transforms physical infrastructure into consumable services.

4.3 HCS Architecture and Main Services

HCS provides separate services for different infrastructure functions. For compute, the studied services included Elastic Cloud Server, Bare Metal Server, image management, and auto-scaling. For storage, the main services studied were Elastic Volume Service for block storage, Scalable File Service for file storage, and Object Storage Service for object storage. For networking, I studied Virtual Private Cloud, Elastic IP, Elastic Load Balance, VPN, Direct Connect, Cloud Connect, and related enterprise networking capabilities. For containers, Cloud Container Engine provides Kubernetes-based container management, while the Software Repository for Container provides container image storage. The service model therefore resembles a complete cloud platform rather than a simple virtualization environment.

4.4 VMware vSphere

VMware vSphere provides a different implementation of a similar infrastructure objective. The core virtualization component is ESXi, which runs directly on the underlying server hardware and hosts virtual machines. vCenter Server provides centralized management of ESXi hosts. Several hosts can be organized into clusters, allowing resources and availability features to be managed at cluster level. During the internship I studied HA, vSphere Standard Switches, vSphere Distributed Switches, datastores, and the virtual machine lifecycle with a focus on centralized deployment, like in the case of configuration management, instead of configuring each server with a certain version and to restrict allowed versions to avoid cnoflicts or any issues, vSphere Lifecycle Manager and Configuration Profiles can be used to reduce configuration drift across hosts.

4.5 HCS vs VMware vSphere

The main similarity between HCS and VMware vSphere is their ability to abstract physical infrastructure into managed resources; compute, networking, storage, virtual machines, management, and availability. The difference is in their scope and abstraction model. HCS is a cloud platform offering a large collection of services, including compute, storage, networking, security, containers, databases, application services, and management. vSphere is primarily a virtualization and infrastructure-management platform. Additional VMware products can extend it into a broader cloud platform to replicate services like the ones provided by HCS. Therefore, HCS was useful during the internship for understanding the cloud-service model, while vSphere was particularly suitable for the practical laboratory because it allowed practical study of the ose concepts. The comparison also helped reinforce an important principle: different vendors may implement similar business

capabilities using different architectures and technologies, the important thing is to know wehen to use which depending on the business goals and requirements, which again highlights the importance of the project methodology study that was done in the beginning of the project.

4.6 Networking: Virtual Switching & Routing

Initially, when trying to understand and connect the cloud and virtualization concepts on top of my pre-acquired networking knowledge, so many things didn't add up, which cause a real bottleneck for learning and implementing. So an understanding from the ground up of virtualized networking independently from the previous networking concepts. And the saying goes; teaching someone who knows nothing about the topic is far easier than someone who has some rooted misconceptions built on similar yet different aspects. These new concepts included:

4.6.1 Interface Classification & System Management

- **Management Interface (Management1):** Dedicated out-of-band interface bound to the management VMkernel/network. Used exclusively for control plane, remote administration, and automation interfaces, including eAPI, SSH, and Web UI.
- **Console Access:** Managed via the virtual serial port, decoupling management plane operations from network state.
- **Data Interfaces (Ethernet1..N):** Virtualized switching/routing ports. Network adapters provisioned at the hypervisor level map 1:1 to EOS interfaces. Interface MAC addresses are generated dynamically during initial system boot.

4.6.2 Switching & Routing

1. Layer 2 Trunking: Forwards traffic across multiple VLANs between network elements over a single link.
2. L3 switchs can operate in switching or routing modes (no switchport) Switch Virtual Interfaces (SVI): Serves as a virtual Layer 3 interface bound to a specific VLAN, operating as the default gateway for workloads within that VLAN's subnet.
3. Inter-VLAN Routing: Inter-VLAN packet forwarding requires explicit global enablement of the IP routing engine.

4.6.3 Network Address Translation (NAT) Implementation

NAT on Arista EOS, the L3 vSwitch used in this project requires an Access Control List (ACL) traffic filter to identify flows before applying translation rules. Under Source NAT Overload (Many-to-One), multiple private IP endpoints map to a single public IP address using transport-layer port multiplexing.

4.7 Virtualization

- **.vmdk:** Virtual Disk file format consisting of a text descriptor header file (defining geometry, adapter types, and storage parameters) and a raw binary payload file containing OS files and app data.
- **.vmx:** The XML/text-based configuration file containing virtual hardware specifications.
- **.iso:** Image payload containing installation scripts and bootloader files used to provision operating systems onto .vmdk disks.

Some Features learnt:

- ESXi does not permanently lock down physical RAM assigned to a VM. Physical memory is managed dynamically. If a host faces memory contention and a VM demands its allocated allocation, ESXi leverages hypervisor swapping (storing inactive VM memory pages onto backing disk storage) to free up physical host RAM.
- Dynamic Resource Scheduling (DRS): Continuously monitors host CPU/RAM utilization across a cluster and performs live migrations (vMotion) of active VMs to underutilized hosts with zero service interruption (Requires shared storage).
- High Availability (HA): Automatically detects physical host hardware failures and restarts affected VMs on healthy cluster nodes (Requires shared storage).

4.8 Hardware Security & Attestation Architecture

ESXi implements a hardware-backed security mechanism to ensure platform integrity during system startup:

1. **TPM 2.0:** Secure hardware vault storing cryptographic secrets, keys, and digital certificates.
2. **Intel TXT:** Cryptographically measures all software execution modules during boot against known-good baseline hashes stored in the TPM.
3. **Secure Boot & Attestation:** Validates signatures of boot files and continuously measures runtime kernel file integrity to prevent rootkits or compromised bootloaders from executing.

4.9 Virtual Switching & Host Packet Flow Mechanics

4.9.1 VMware Host Network Traffic Forwarding

Understanding network operations requires evaluating how packets move across the physical NIC (vmnic), virtual switches (vSS/vDS), and host/guest IP stacks.

Forged Transmits: Controls whether a VM can send outgoing frames with a source MAC address that does not match its assigned virtual NIC address. Setting this to Reject prevents MAC spoofing attacks within the virtual network.

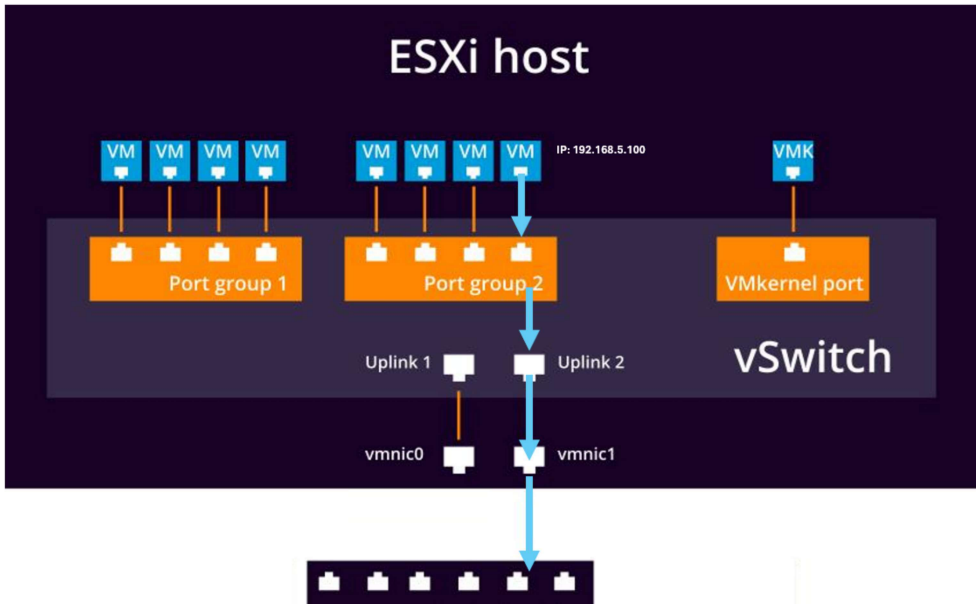


Figure 4.2: ESXi internal virtualized networking

4.9.2 Standard Switch (vSS) vs. Distributed Switch (vDS)

- **vStandard Switch (vSS):** Configured and managed locally on an individual host. Required for core host boot management interfaces.
- **vDistributed Switch (vDS):** Centralized control plane managed via vCenter across an entire host cluster, simplifying switch management.

4.10 Spine-Leaf Network Architecture

Modern data center networks replace traditional hierarchical tree designs with a two-tier Spine-Leaf architecture.

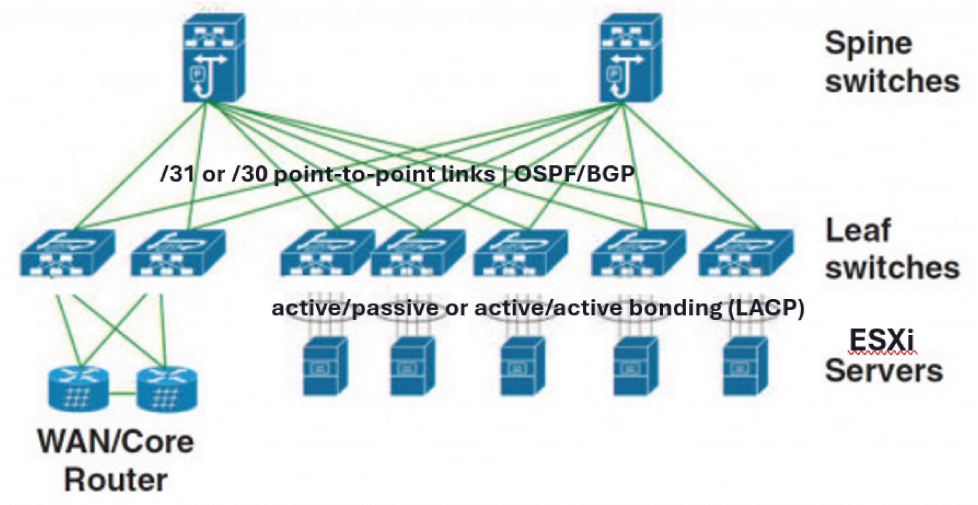


Figure 4.3: Spine Leaf Network Architecture Diagram

- **Leaf Switches (Top-of-Rack / ToR):** Connect endpoints (servers, storage arrays, firewalls). Topologies range from single-homed connections to dual-homed multi-chassis link aggregation (MLAG) configurations.
- **Spine Switches:** Form the high-speed backbone core. Every leaf switch connects to every spine switch, maintaining equidistant single-hop latency across workloads.
- **TCP Incast Mitigation:** Deep buffer switches are placed at leaf nodes to absorb simultaneous packet bursts sent by storage clusters to a single host node, preventing packet loss caused by buffer exhaustion.

4.11 Linux System Storage Management & Container Platforms

4.11.1 Storage Partitioning Behaviour in Enterprise Distributions

Enterprise Linux installers (such as Ubuntu Server) default to conservative volume management policies during initial setup. By default, the installer provisions only 50% of available disk space (up to 100 GB) to the root Logical Volume (/dev/ubuntu-vg/ubuntu-lv), reserving the remaining capacity for snapshot space, system recovery, and manual volume expansions. Administrators must manually extend the Volume Group (VG) and Logical Volume (LV) to utilize the entire storage array.

4.11.2 Kubernetes Platform Architecture Overview

Kubernetes orchestrates containerized workloads across a distributed cluster by separating control plane components from worker node runtime engines.

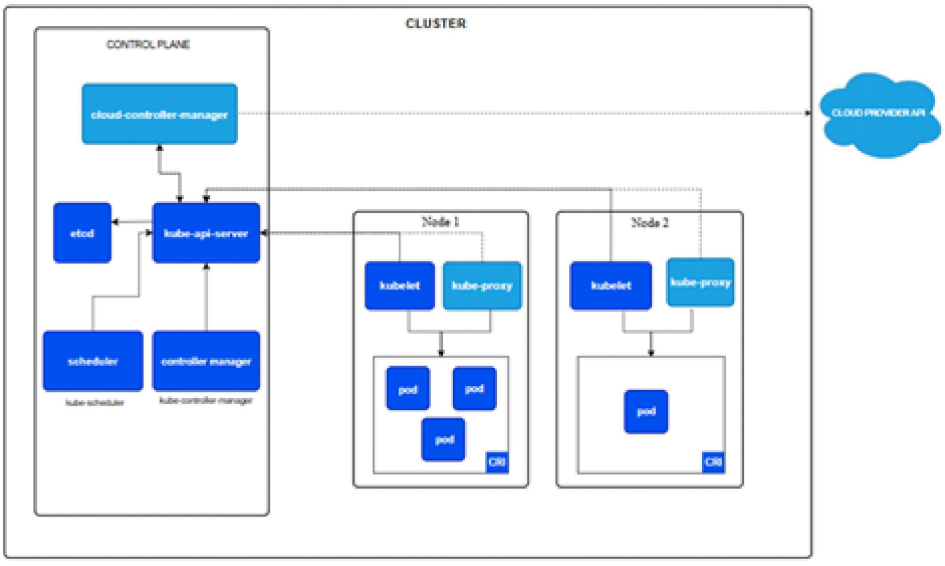


Figure 4.4: Kubernetes components diagram

- **Control Plane Components:**
  - kube-apiserver: Exposes the REST API and serves as the central orchestration front-end.
  - etcd: Distributed, consistent key-value store maintaining cluster state records.
  - kube-scheduler: Assigns unscheduled Pods to worker nodes based on resource constraints.
  - kube-controller-manager: Runs controller loops managing node lifecycle, replication, and endpoints.
- **Worker Node Components:**
  - kubelet: Agent enforcing pod specifications provided by the control plane.
  - kube-proxy: Manages network routing rules and load balancing across cluster pods.
  - Container Runtime Engine (e.g., containerd): Handles image pulling and runs the isolated container instances.

Chapter 5

System Design and Architecture

The implementation started with the infrastructure layer because every subsequent component depended on it. VMware ESXi and vCenter Server were deployed first, followed by the virtualized network and the vSphere cluster configuration. Once the infrastructure was sufficiently stable, Kubernetes was deployed on top of the VMware environment. Finally, high-availability mechanisms were configured at both infrastructure and Kubernetes levels through vSphere HA and pod anti-affinity. This order directly followed the dependencies identified during the design phase. It also made troubleshooting more manageable because each layer could be validated before being used as a foundation for the next one. The following chapters therefore move from the theoretical and architectural study toward the concrete realization of the designed infrastructure.

Due to lack of physical resources to replicate this physical data center architecture, as the only resources are my personal laptop and a borrowed laptop. So since the practical environment was built under significant hardware constraints I had to Virtualize the whole thing, and since my ultimate goal was realism, I tried to understand the components of the enterprise datacenter and map them to my environment. Instead of using a full enterprise data center, the infrastructure was reproduced using available computers and nested virtualization. This created realistic engineering difficulties. The available network adapter was not directly compatible with the standard ESXi installation image, and the workstation configuration had to be modified to allow nested virtualization.

Several ESXi servers are typically deployed in a datacenter, for example 10 like in the image, in my virtualization approach, 9 of them were deployed as VMs inside VMware workstation pro on my windows machine, which is a virtualization platform, and one is deployed as an actual physical server on the second laptop. (The color coding between the two images can help associate the virtualized components with their actual physical counter parts.)

So the physical infrastructure of the project becomes as follows:

5.1 Logical Design

The infrastructure follows a layered architecture, where each layer is responsible for a distinct function and exposes services to the layer above it.

The ESXi hosts are managed centrally by a single VMware vCenter Server Appliance (VCSA) deployed as a virtual machine

At the lowest level is the physical infrastructure consisting of servers, storage, and network connectivity. Above this layer is VMware ESXi, which provides hardware virtu-

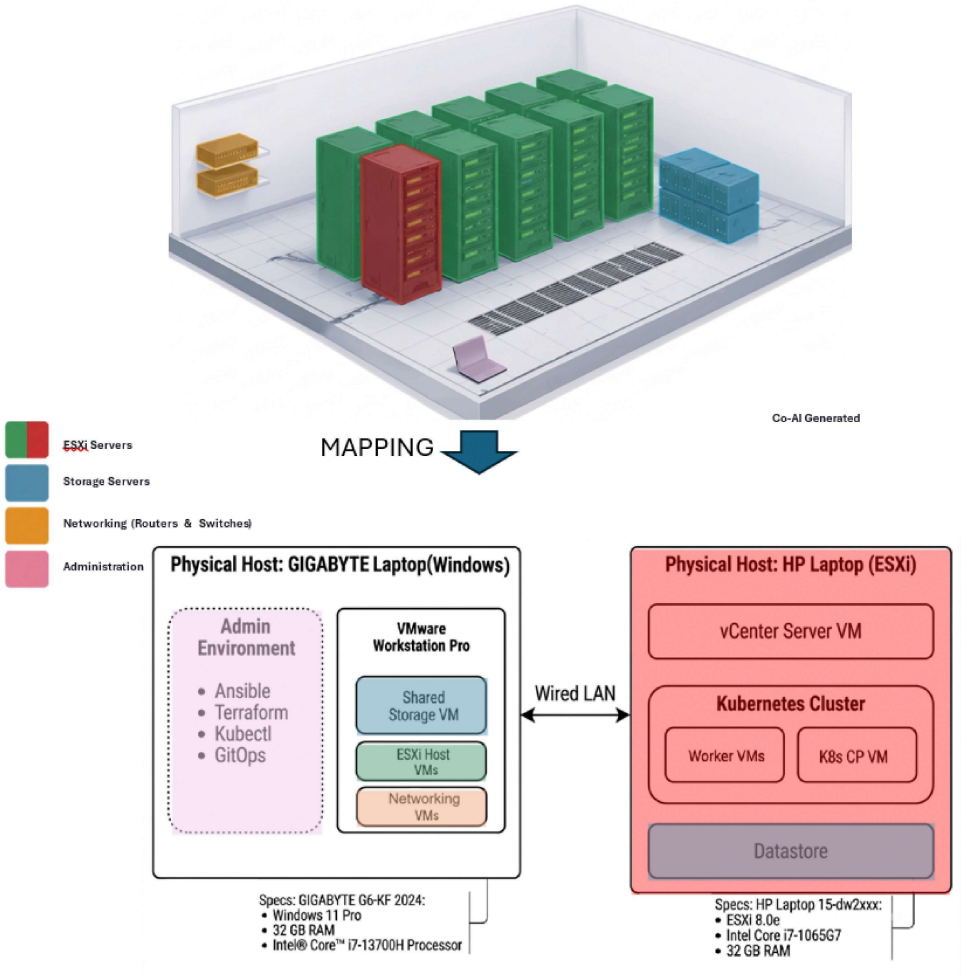


Figure 5.1: Physical to virtual data center component mapping

alization. vCenter Server provides centralized management, while ESXi hosts are grouped into a vSphere cluster. The network layer uses virtual switches and a laboratory datacenter topology based on a spine-leaf model. Kubernetes is deployed on virtual machines hosted by the VMware environment. Finally, the application layer consists of workloads such as MySQL and PostgreSQL, together with Kubernetes Services and Ingress resources. This layered design reflects the same principle encountered when studying HCS: physical resources are progressively abstracted until they can be consumed by applications as services.

5.2 Compute and Virtualization Layer

The ESXi hypervisor provides the compute virtualization layer. One important concept studied during the internship was that allocating memory to a virtual machine does not necessarily mean permanently reserving an equivalent quantity of physical memory in a simple one-to-one manner. ESXi manages physical resources dynamically according to workload requirements.

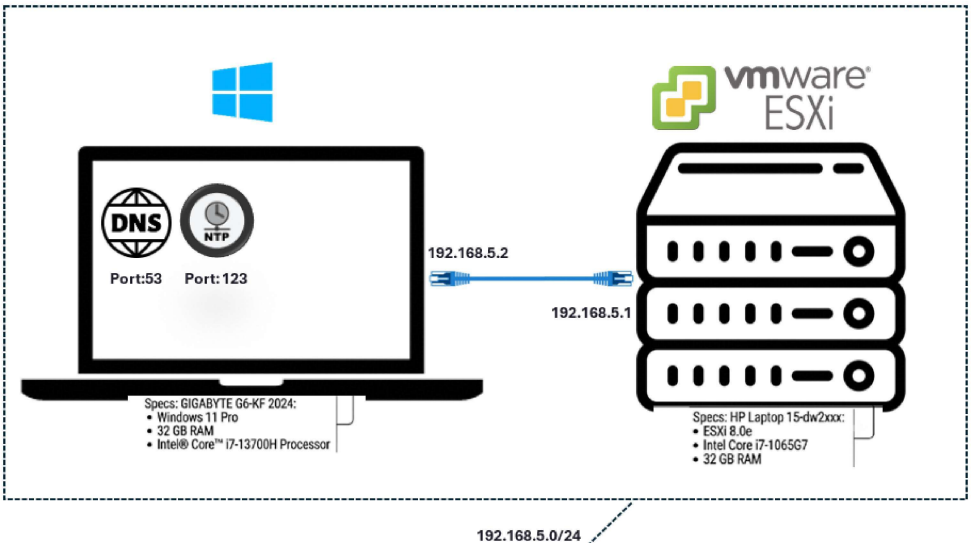


Figure 5.2: Physical infrastructure and host connectivity diagram

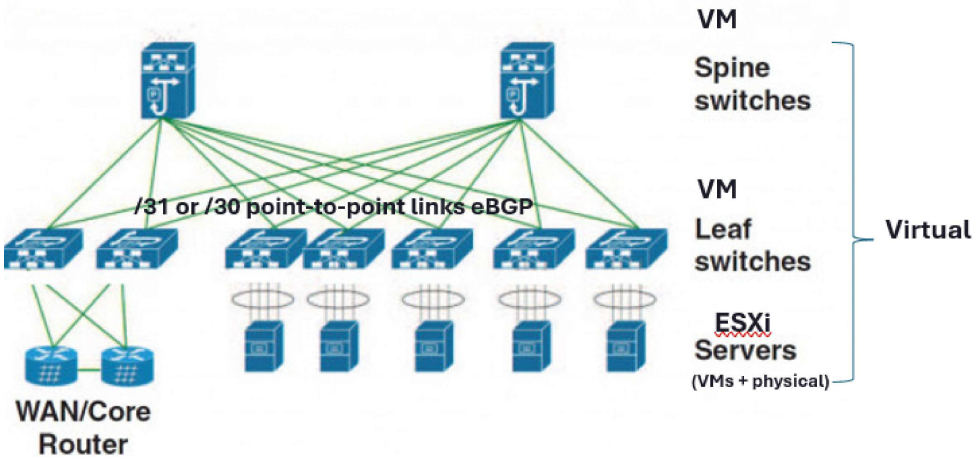


Figure 5.3: Physical Network Design

5.3 Network Architecture

The network design was based on a data-center-oriented spine-leaf model. Traditional three-tier networks were originally designed largely around north-south traffic patterns. Spine-leaf networks are well suited to environments with significant east-west traffic because leaf switches can reach other leaves through multiple spine paths. The underlay was designed as a layer-3 network to avoid very large layer-2 broadcast domains and to provide predictable routing. The main concepts studied included BGP, ECMP, VLANs, loopback addresses and VXLAN. BGP was studied as a policy-based routing protocol and as a common protocol for data-center leaf-spine deployments. The notes also emphasized the use of ECMP to distribute traffic between multiple spine paths and the use of loopback addresses as stable identifiers for network devices.

the Virtualized Network connecting the VMs is to simulate the data center's traffic patterns (East-West and North-South traffic): The underlay network linking the esxi hosts is in place, the deployed VMs will ultimately use that network architecture but will also have their own logical architecture, many options were presented like VXLAN/Geneve overlay, but considering the time and resources, the simplest approach was using plain-old

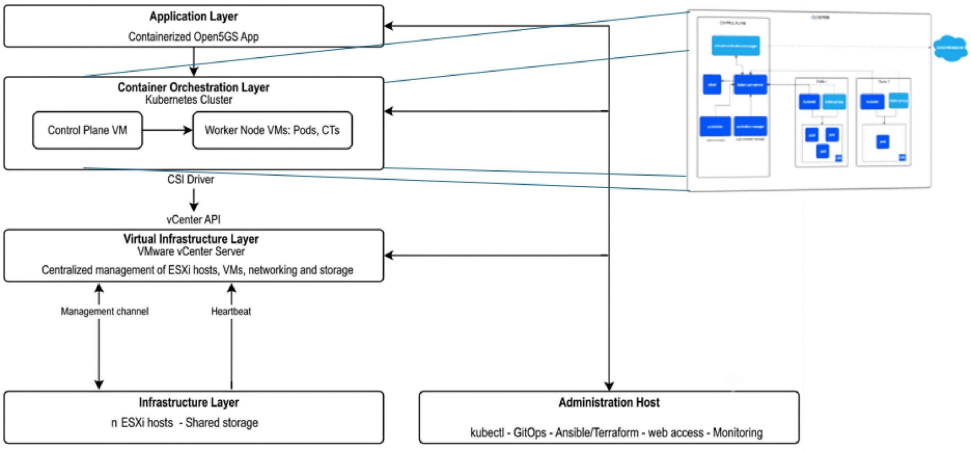


Figure 5.4: Logical diagram

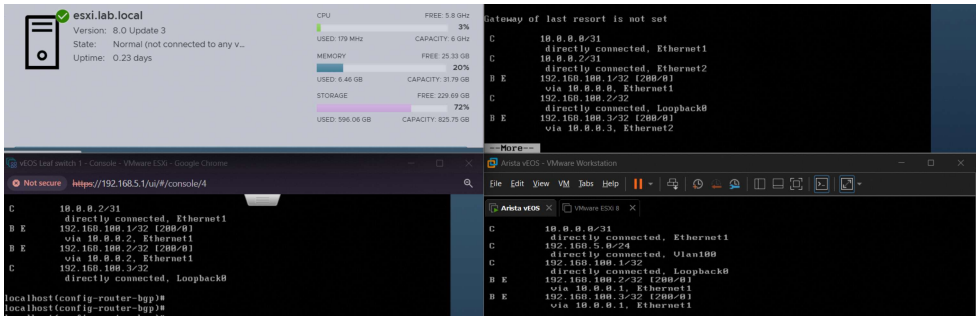


Figure 5.5: Virtualized network connectivity and routing validation

VLANs. ()

The VM sends traffic from its vNIC's interface with the destination VM's IP to the vSwitch which encapsulates the traffic in the VLAN frame and routes it to the leaf switch, which uses BGP/OSPF to route the traffic using one of the spine switches, which in turn uses the same protocols to send it to the destination leaf switch and then to the vSwitch in the destination esxi host which decapsulates the packets and forwards them to the destination VM.

5.4 Kubernetes Workload Layer

Kubernetes was selected as the workload-management platform. The cluster provides the abstraction required to deploy applications independently of the specific underlying virtual machines. The project focused on several Kubernetes resources: pods, services, PersistentVolumes, PersistentVolumeClaims, StorageClasses, Ingress resources, and scheduling constraints. The important architectural principle was to keep the Kubernetes layer independent from direct management of VMware resources. CSI provides the storage integration point, while Kubernetes scheduling manages application placement.

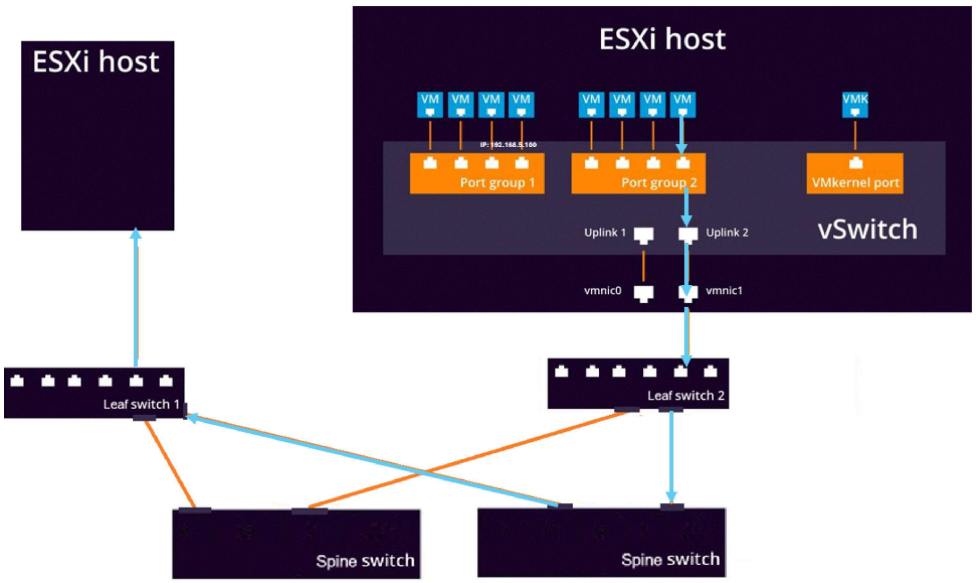


Figure 5.6: Example traffic flow between VMs in different ESXi hosts

Chapter 6

Implementation of the Infrastructure

6.1 Laboratory Environment

These hardware limitations influenced the design of the virtual network and the number of hosts that could be reproduced, and introduced major difficulties that were impossible to predict in the project management phase because allocated timeframes for the tasks were refreneced from sources that worked on real deployments. The Deployment was as follows:

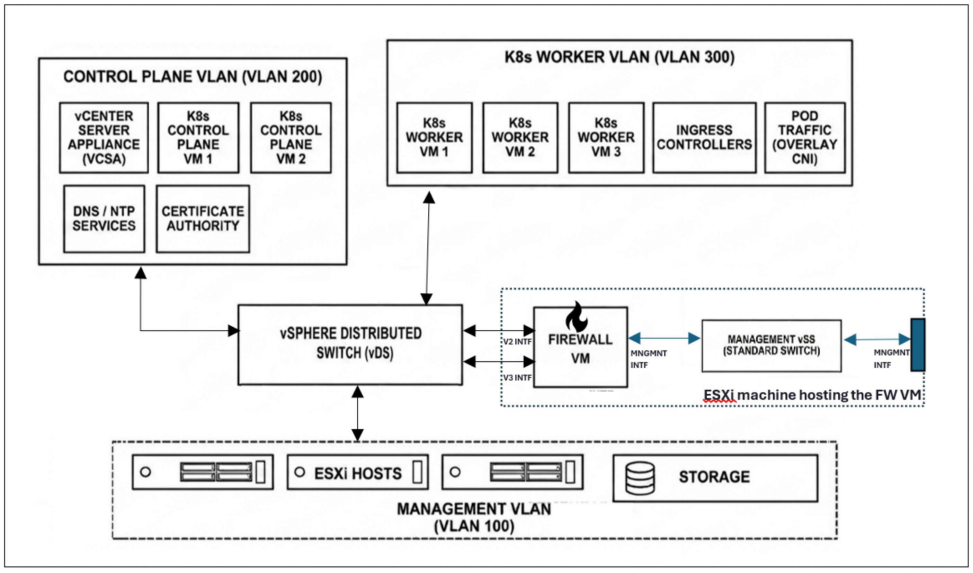


Figure 6.1: Laboratory workload and management network design

6.2 ESXi Installation

The first ESXi installation required solving a hardware-compatibility issue. The standard ESXi image did not include a suitable driver for the available network adapter. A customized installation image was therefore prepared using community-supported drivers. After creating the installation media, ESXi was installed and its management interface was accessed to perform the initial configuration. The first successful access to the ESXi management interface provided the basis for the rest of the VMware deployment.

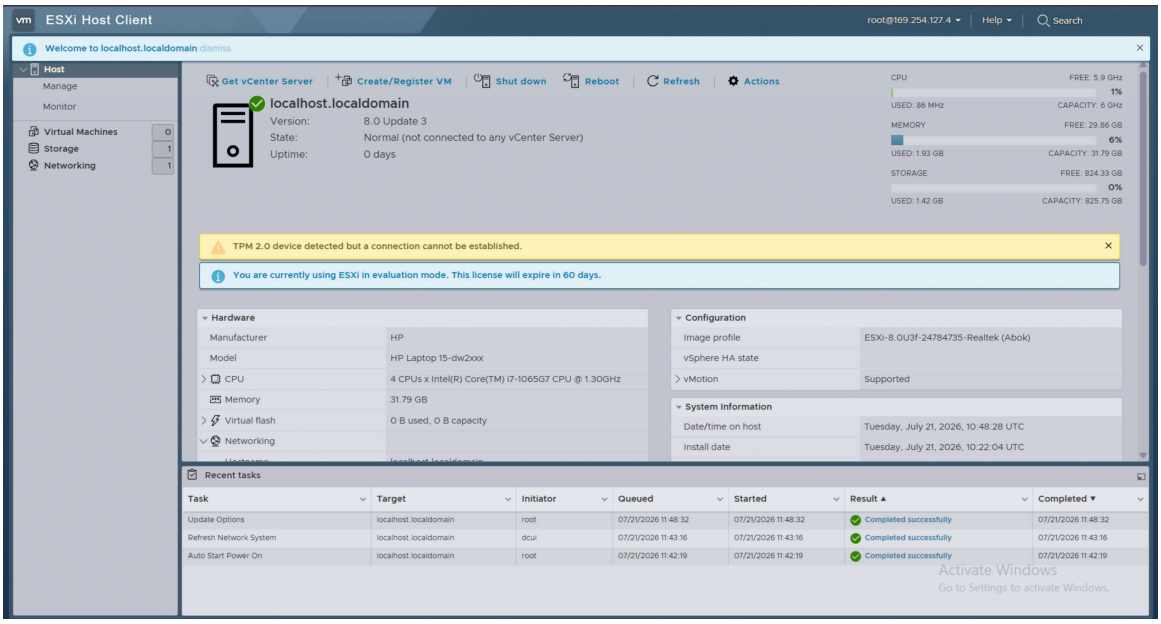


Figure 6.2: ESXi Host Client after installation

6.3 vCenter Server Deployment

Once ESXi was available, vCenter Server was deployed as the central management component. The deployment involved creating the required virtual machine and installing the vCenter software. The process was not straightforward because the laboratory hardware introduced resource and compatibility constraints. Several deployment attempts were required, including configuration changes and troubleshooting. This was a useful practical example of the difference between a documented deployment procedure and an actual infrastructure deployment where hardware and environmental constraints can affect the result.

6.4 vSphere Datacenter and Cluster Configuration

After vCenter became operational, a datacenter object was created followed by a vSphere cluster. The cluster was configured with HA and DRS. I studied the distinction between a datacenter object and a cluster. The datacenter provides an organizational container, while the cluster groups ESXi hosts into a managed resource and availability domain. The host-joining process also demonstrated the importance of maintenance mode and management-plane dependencies.

6.5 Nested ESXi Environment

Because the laboratory had limited physical hardware, additional ESXi hosts were created as virtual machines in VMware Workstation Pro. Nested virtualization required several changes to the Windows host environment. Hyper-V-related features and other virtualization mechanisms had to be disabled, while CPU virtualization had to be exposed to the nested virtual machines. The virtual machines were configured in bridged networking mode to allow them to communicate with the rest of the laboratory network. This phase

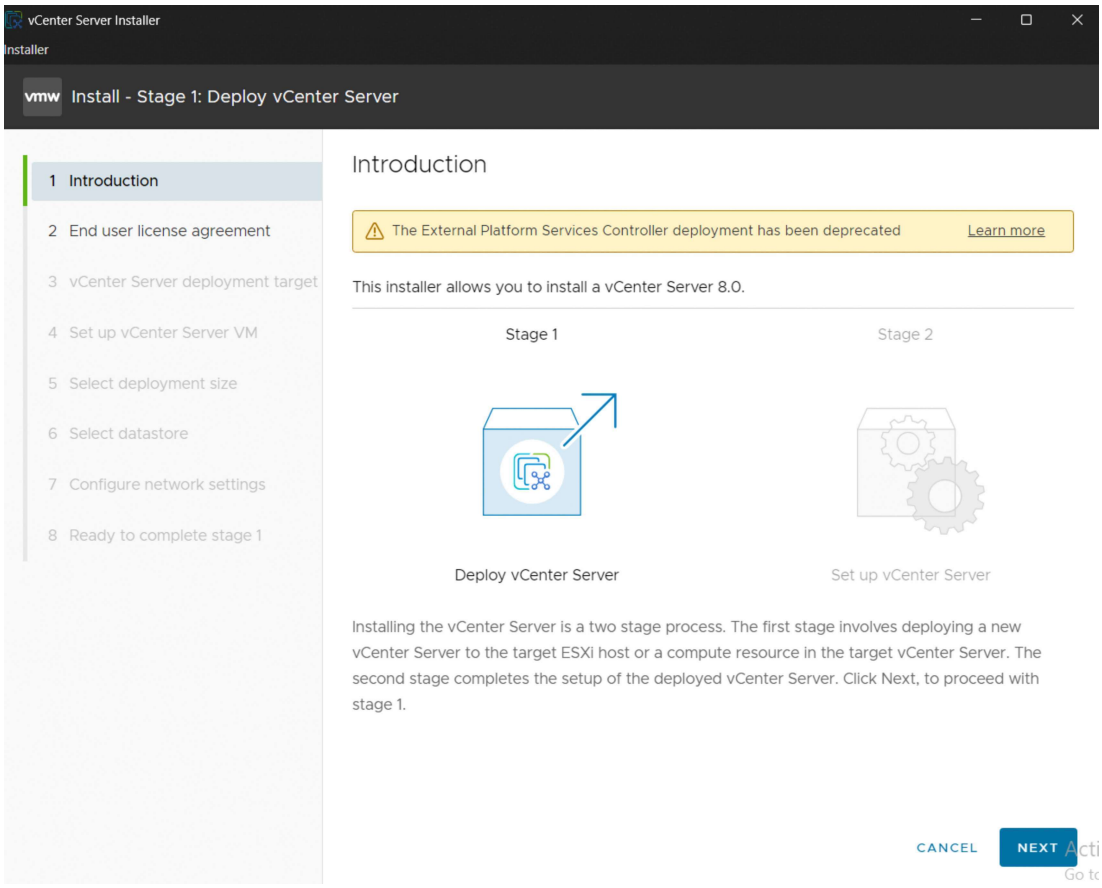


Figure 6.3: vCenter Server deployment installer

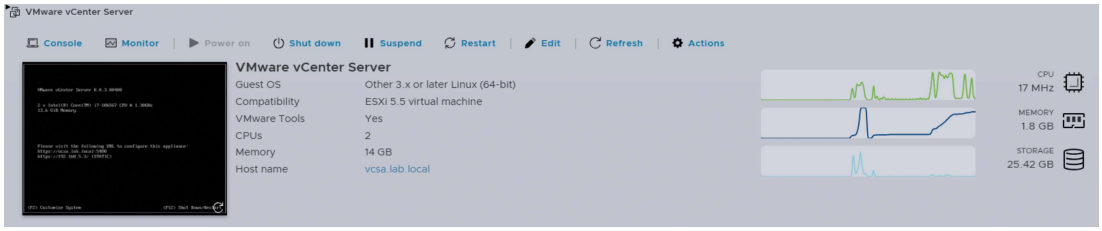


Figure 6.4: Deployed vCenter Server virtual machine

was particularly useful for understanding the difference between the physical host network adapter, the VMware Workstation network, the ESXi virtual switch, and the guest operating system network interface.

6.6 Virtual Data-Center Network

An Arista vEOS virtual switch was deployed to reproduce the behavior of a layer-3 data-center switch. The switch was built from its installation media and virtual disk rather than simply relying on an imported appliance. The network topology was designed around leaf and spine components. Connections were configured one at a time, and connectivity was verified before moving to the next link. This reduced the troubleshooting scope and allowed problems to be isolated. Specific virtual-switch settings were also adjusted to support the nested network architecture, including promiscuous mode and forged-transmit

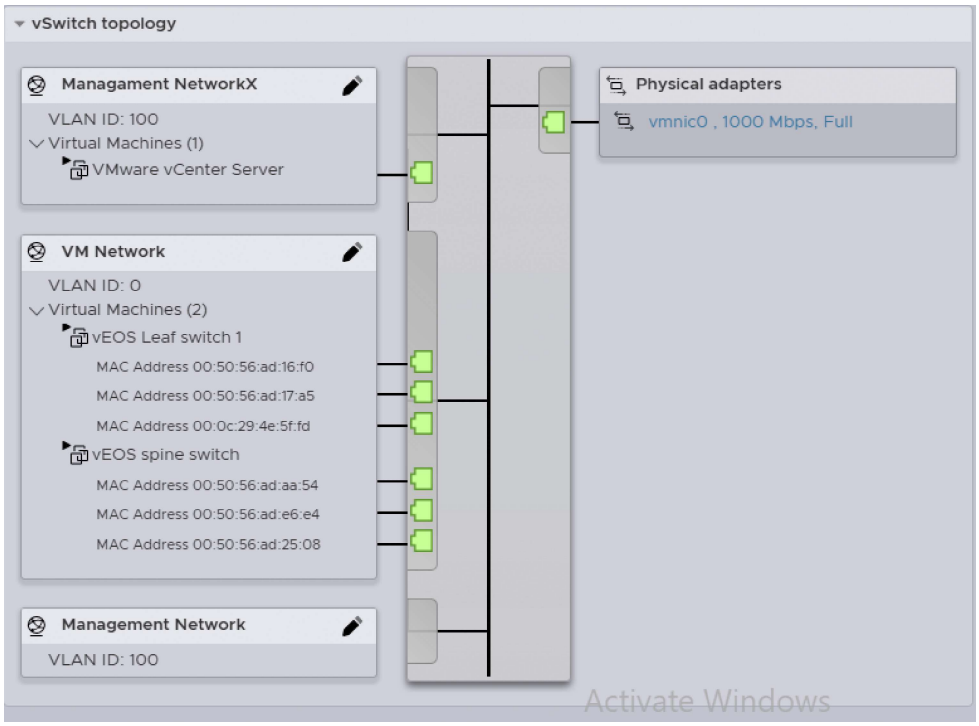


Figure 6.5: Connectivity between the workstation and the ESXi environment

behavior. The final topology provided the network foundation on which the Kubernetes environment could operate.

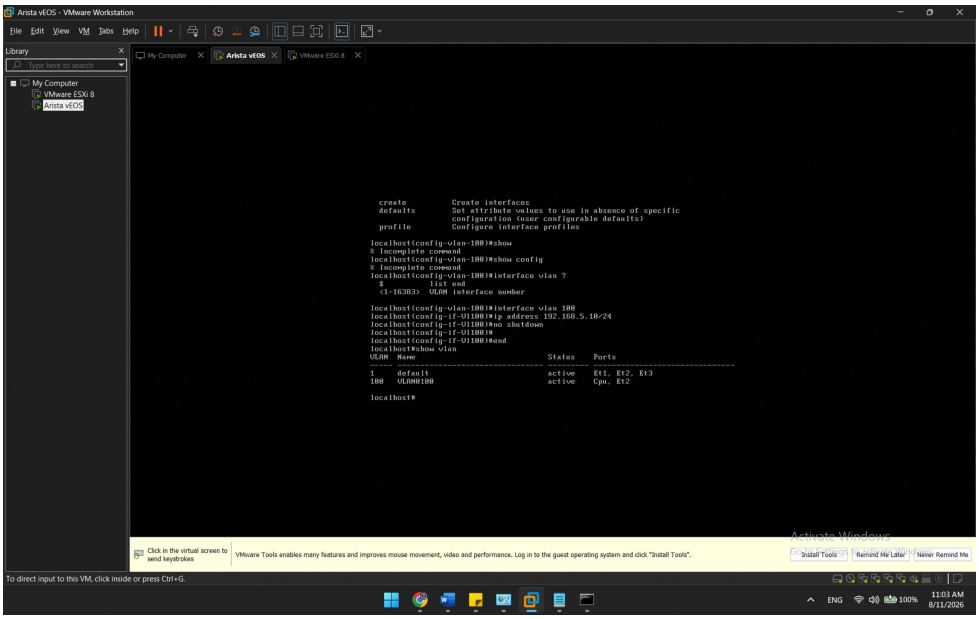


Figure 6.6: Virtual switch and VLAN configuration

Chapter 7

High Availability and Workload Distribution

7.1 Infrastructure-Level High Availability

High availability is not achieved by adding replicas alone. A workload may have multiple virtual machines while still depending on a single physical server, storage system, or network component. Therefore, the project considered availability at both the infrastructure and application levels.

7.2 vSphere HA

vSphere HA was enabled on the VMware cluster. The objective is to allow virtual machines running on a failed ESXi host to be restarted on another available host. The practical value of HA therefore depends on the existence of other hosts and appropriate shared or accessible storage. The feature was studied as the infrastructure-level mechanism protecting Kubernetes virtual machines.

7.3 Pod Anti-Affinity

Pod anti-affinity was configured to prevent related workloads from being concentrated on the same Kubernetes node. This is important for replicated applications because placing all replicas on one node creates a single point of failure. The configuration was tested by deploying multiple replicas and observing their placement across the available nodes.

Chapter 8

Validation and Results

8.1 Validation Strategy

The validation process followed the same layered approach used during implementation. Each major subsystem was first tested independently and then tested as part of the complete architecture. This prevented a failure in one layer from being incorrectly attributed to another component. The principal validation areas were storage provisioning, data persistence, external networking, ingress routing, and high availability.

8.2 Network Validation

The virtual network was tested progressively from the ESXi host through the leaf and spine layers. Connectivity was verified after each stage. The Kubernetes services were then exposed and tested externally. This confirmed that the virtualization network could support the higher-level Kubernetes networking requirements.

8.3 Availability Validation

The availability configuration was evaluated through the combination of vSphere HA and Kubernetes anti-affinity. The placement of replicas across Kubernetes nodes was checked. The relationship between Kubernetes nodes and the underlying ESXi hosts was also considered to ensure that workload distribution was meaningful from a physical-failure perspective.

Chapter 9

Critical Analysis

9.1 Technical Challenges

The first major challenge was hardware compatibility. The available computer's network adapter was not supported directly by the standard ESXi image. Solving this issue required understanding ESXi installation images and driver packaging. The second challenge was nested virtualization. Running ESXi inside VMware Workstation required changes to the Windows host and careful configuration of CPU virtualization. The third challenge was vCenter deployment. Resource limitations and compatibility issues required several deployment attempts and configuration changes. The fourth challenge was virtual networking. The laboratory had to reproduce behavior normally implemented by physical data-center switches, while traffic passed through several layers of virtualization. These difficulties demonstrated that infrastructure engineering involves much more than installing software according to documentation.

9.2 Lessons Learned

The first major lesson was the importance of understanding architecture rather than memorizing commands. When a configuration failed, understanding the relationships between the components was much more useful than repeating the installation procedure. The second lesson was the importance of progressive troubleshooting. Connecting and testing each network component individually proved more efficient than deploying the full topology and attempting to diagnose everything afterward. The third lesson concerned abstraction. Cloud platforms, VMware, and Kubernetes all provide different abstraction levels. Understanding the boundary between those levels is essential when troubleshooting. The fourth lesson was that availability must be considered across failure domains rather than only at the application level.

9.3 Project Management Lessons

The internship also provided experience in project management. The structured approach from requirements to design, implementation, testing, and documentation was useful because the project contained several dependencies. Documentation was particularly important because the project included many different technologies and configuration decisions. I also learned that engineers should understand the business objective behind a technical

solution. The project therefore reinforced the idea that an engineer should be able not only to execute a configuration but also to explain why the configuration is required and how it contributes to the final objective.

9.4 Recommendations and Perspectives

A production implementation should use dedicated physical ESXi hosts rather than rely on nested virtualization. The network infrastructure should use redundant physical links and devices, with appropriate routing and failure detection. Infrastructure deployment should be automated through Infrastructure as Code and configuration-management tools. Monitoring and centralized logging should be integrated from the beginning rather than added after deployment. The storage architecture should also include backup and restore mechanisms, because persistence alone does not provide complete data protection. Security should be treated as a cross-layer concern covering hardware management, virtualization, Kubernetes permissions, network policies, credentials, secrets, and application exposure. Finally, future work could extend the project toward an automated private-cloud platform combining VMware infrastructure, Kubernetes, Infrastructure as Code, monitoring, security, and application deployment.

Chapter 10

General Conclusion

The objective of this internship was to understand how enterprise infrastructure can support cloud-native workloads through virtualization, storage integration, networking, and high availability. The project began with a study of Huawei Cloud Stack in order to understand the architecture and service model of a private-cloud platform. This study provided a broader view of how compute, storage, networking, security, management, and application services can be combined into a cloud environment. The practical work then focused on VMware vSphere as an alternative implementation of the same general infrastructure objective. A VMware laboratory environment was built using ESXi, vCenter Server, clusters, virtual networking, and nested virtualization. A data-center-oriented network was reproduced using a virtual spine-leaf topology. Kubernetes was subsequently deployed above the VMware layer.

The high-availability objective was addressed at two levels. vSphere HA provided infrastructure-level virtual-machine recovery, while Kubernetes pod anti-affinity was used to distribute application workloads across different nodes. The internship also involved significant troubleshooting. Hardware incompatibility, nested virtualization, vCenter deployment, and virtual networking required practical investigation and iterative configuration. Although the laboratory environment did not reproduce the scale or redundancy of a production data center, it provided a useful proof of concept and, more importantly, a practical understanding of how the different infrastructure layers interact. The project therefore achieved its educational and technical objectives. It provided experience not only with VMware and Kubernetes, but also with architecture, networking, storage, high availability, troubleshooting, documentation, and project management. The natural continuation of this work would be to move toward a more production-oriented platform based on dedicated hardware, automated provisioning, centralized monitoring, stronger security controls, and broader security measures and best practices.

SUP'COM
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Ecole Supérieure des Communications de Tunis
Higher School of Communications of Tunis

Ecole Supérieure des Communications de Tunis

Journal de Stage

Type de stage	☐ Stage de formation humaine	■ Stage Ingénieur
Nom et prénom de l'élève ingénieur	Mohamed Amin AYED	
Etablissement d'accueil	HUAWEI	
Période du stage	de 01/07/2026 à 31/08/2026 *(Minimum 5 semaines)*	

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2083 Cité Technologique des Communications – El Ghazala - Ariana, Tunisie
– http://www.supcom.mincom.tn - E-mail : direction.stages@supcom.tn

Fiche identification du stage

Identification de l'établissement d'accueil				
Nom de l'entreprise	HUAWEI Technologies Tunis S.A.R.L			
Domaine d'activités	ICT			
Adresse du lieu du stage	Immeuble Fredj rue feuille de l'erable 1053 *CP/Ville :Tunis* *Pays :* Tunisie			
Responsable administratif d'accueil des stagiaires	*Nom*	Hana El Karou	*Fonction*	HR assistant
	N° téléphone	54603784	*N° fax*	……..
	e-mail	Hana.el.karou1@huawei.com		
Site web	……			

Service bénéficiaire du stage				
Nom du service	Tunis Office Delivery & Service Business Dept			
Encadrant du stage	*Nom*	Mohamed Ali Boujelben	*Fonction*	Fulfillment Responsible
	N° téléphone	……..	*N° fax*	……..
	e-mail	mohamed.ali.boujelben@huawei.com		

Tuteur à SUP'COM	……..

Identification de l'offre de stage	
Période du stage	de 01/07/2026 à 31/08/2026 *(1 à 2 mois durant juillet–août)*
Programme du stage *(joindre un descriptif complémentaire si c'est nécessaire)*	Cloud deployment on VMware vSphere

Réglementation des stages

1. Définition des stages techniques

Les stages techniques constituent un complément de formation pratique obligatoire aux élèves ingénieurs de SUP'COM. Ils sont effectués dans des établissements d'accueil (entreprises, associations, laboratoires universitaires, …) sous la responsabilité d'un ou plusieurs encadrants. Au cours de leurs scolarités les élèves-ingénieurs sont astreints à effectuer les deux stages techniques suivants :

1.1) Stage de formation humaine

Ce premier stage est effectué au début du troisième semestre. Il a pour objectif de permettre à chaque élève ingénieur de faire l'expérience de la vie en entreprise, et plus largement de la vie professionnelle dans le but d'enrichir son expérience sociotechnique.

Ce stage peut prendre des formes variées :

- Stages en entreprise qui permettent à l'élève d'observer et de comprendre sa structuration, ses moyens et ses méthodes de travail. Après la découverte de l'entreprise, le stagiaire peut participer à des travaux d'application ou d'ingénierie.
- Contribution à des activités techniques ou sociales au sein d'une association ou d'une organisation humanitaire.

1.2) Stage Ingénieur

Ce deuxième stage est effectué au début du cinquième semestre. C'est un stage de perfectionnement dans l'entreprise. Il permettra à l'élève de mettre en œuvre les connaissances acquises durant sa scolarité pour conduire, présenter et mener à bien une solution pour répondre à la demande d'une entreprise. Ce stage a pour objectif pour l'élève ingénieur d'/de :

- Appréhender la vie professionnelle et le monde de l'entreprise en s'intégrant dans une équipe de travail.
- Apprécier, au niveau de l'entreprise, ses ressources (qualités, savoirs, aptitudes), ses potentialités et les aspects à développer.
- Mieux connaître les éléments qui faciliteront ses choix et son orientation d'abord pour son projet de fin d'études puis son premier emploi.

2) Activités et responsabilités de l'élève

2.1) Activités durant le stage

- Accomplissement d'un programme de stage consistant avec le meilleur apport pour le stagiaire.
- Respect des horaires de travail et du règlement interne de l'établissement d'accueil.
- Description dans le journal du stage des tâches journalières effectuées.
- Rédaction d'un rapport couvrant le déroulement du stage dont une copie sera remise avant la fin du stage à l'encadrant dans l'établissement d'accueil et le tuteur à SUP'COM.

2.2) Activités après le stage

** Présenter à la Direction des stages de SUP'COM dans les délais et selon les modalités communiquées :*

- *Un rapport du stage avec la première page portant le cachet de l'établissement d'accueil et les signatures de l'encadrant externe et le tuteur à SUP'COM,*
- *Le journal du stage avec la première page portant le cachet de l'établissement d'accueil et les signatures de l'encadrant externe et éventuellement le tuteur à SUP'COM,*
- *Une copie de l'attestation de fin de stage.*
- *Une évaluation du stage (conditions de déroulement, apports,..)*

** Soutenir son travail de stage devant un jury d'évaluation composé par des enseignants de SUP'COM et un représentant de l'établissement d'accueil du stagiaire.*

3) Suivi du stagiaire

- Le suivi du stagiaire sera assuré conjointement par un encadrant dans l'établissement d'accueil et éventuellement un tuteur à SUP'COM.
- Les encadrants dans l'établissement d'accueil veilleront au bon déroulement du stage en fournissant au stagiaire la documentation et la logistique nécessaires et guideront le stagiaire dans l'accomplissement de son programme de stage.
- Le tuteur à SUP'COM interviendra au moins trois fois : une 1ère rencontre avec le stagiaire avant le démarrage du stage pour bien le préparer, un 2ème contact en milieu de la période du stage pour évaluer l'avancement du stagiaire et lui fournir les recommandations nécessaires et enfin une 3ème rencontre pour guider le stagiaire dans la préparation de son rapport de stage et sa soutenance orale.
- En complément aux moyens de l'établissement d'accueil, SUP'COM mettra à la disposition du stagiaire tout autre moyen de l'école jugé d'un intérêt pour l'établissement d'accueil pour la bonne réussite de l'exécution du programme de stage (accès à la bibliothèque, aux outils et équipements techniques, conseil des enseignants chercheurs,…).
- Compte tenu de l'importance de l'appréciation des encadrants externes dans l'évaluation du stagiaire, ils sont invités à compléter un formulaire d'évaluation du stagiaire qui doit être envoyé d'une manière confidentielle à SUP'COM. Le tuteur à SUP'COM, après coordination avec les responsables de l'établissement d'accueil, peut envisager d'effectuer quelques visites sur site pour contrôler le stagiaire et coordonner le suivi de ses activités de stage avec les encadrants dans l'établissement d'accueil.

4) Consignes générales pour la rédaction du rapport de stage

- Le rapport de stage à rédiger par l'élève ingénieur sera conçu comme une étape de prise de recul par rapport à l'expérience du stage, dans tous ses aspects.
- L'objectif de ce rapport est d'écrire les réflexions et les analyses que cette expérience aura pu susciter après coup.
- Les thèmes abordés sont choisis par l'élève et s'appuient sur ses perceptions, difficultés, surprises qui l'ont interrogé durant le stage (son travail, le travail des autres, les rapports dans le travail, l'intégration, la culture de l'entreprise, etc.).
- Le tuteur du stage à SUP'COM peut solliciter le suivi de la rédaction du rapport par un enseignant de langue à SUP'COM s'il juge que l'élève souffre de difficultés linguistiques.

5) Usage responsable de l'IA :

Dans le cadre de la rédaction du rapport de stage, l'utilisation d'outils d'intelligence artificielle générative peut être admise comme un support d'aide (structuration, reformulation, amélioration linguistique), à condition qu'elle demeure encadrée et transparente.

L'étudiant est tenu de déclarer explicitement tout recours à ces outils, en précisant leur nature et les usages qui en ont été faits.

Le contenu du rapport doit toutefois refléter fidèlement son travail personnel, sa compréhension des missions réalisées et sa capacité d'analyse critique. Toute utilisation de l'IA visant à se substituer à la production intellectuelle de l'étudiant, à générer un contenu non maîtrisé ou à présenter des résultats non authentiques est strictement proscrite. Elle peut justifier une non-validation du stage en question.

Le rapport pourra faire l'objet de vérifications, notamment avant et lors de la soutenance, afin d'évaluer la cohérence entre le contenu présenté et les compétences réellement acquises.

6) Evaluation des activités des stages techniques

- A la fin du stage, l'élève ingénieur présente un rapport décrivant ses activités de stage et l'expose devant un jury de soutenance. Ce jury attribue une note qui tient compte de l'appréciation des encadrants du stage, des informations rapportées dans le journal du stage (visé par l'encadrant dans l'entreprise), de la qualité du rapport, de l'exposé oral et de l'exécution du travail demandé et des résultats obtenus.
- Chaque stage fait l'objet d'une évaluation donnant lieu à une note qui doit être supérieure ou égale à 10/20 pour que le stage soit validé.
- Dans le cas d'un stage non validé le jury d'évaluation jugera du complément de travail à réaliser avant de faire une nouvelle évaluation. Le complément du travail peut être l'une des possibilités suivantes : correction du rapport, complément de stage en entreprise, un nouveau stage en entreprise.

Activités journalières du stagiaire

Semaine 1

Journée	**Date** (jj/mm)	**Résumé des activités**
Lundi	29/06	I met with HR, my supervisor, and members of the engineering teams to understand the internship environment and expectations. I completed the setup of my computer and received the information needed to start working. I became familiar with the company's internal policies, the office environment, and the different engineering teams. I also observed the multicultural working environment and how people from different backgrounds cooperate on technical projects.
Mardi	30/06	I started reviewing the technical documentation provided after the initial meetings. The documentation introduced Huawei Cloud Stack as well as general cloud computing concepts. I organized the information into personal notes so that I could easily return to important subjects later. I focused on understanding the general concepts instead of learning only Huawei-specific terminology.
Mercredi	01/07	I continued studying cloud infrastructure and the main services used in cloud environments. I reviewed different HCS deployment scenarios and tried to understand the role of each infrastructure component. I compared the concepts with technologies I had already studied during my academic work. I also started writing questions about subjects that were still unclear to me.
Jeudi	02/07	I continued my research on cloud concepts and HCS deployments. I studied how cloud infrastructure is divided between computing, networking, storage, and management components. I organized my notes into separate subjects to make them easier to use later. The main challenge was the large amount of information that had to be understood in a short period.

Vendredi	03/07	I continued studying the documentation and focused on different HCS services and deployment models. I tried to understand how the same cloud concepts can be adapted to different customer requirements. I also reviewed the relationship between infrastructure resources and the services running on top of them. This helped me build a stronger foundation before moving to the project-specific technologies.
Samedi	04/07	I reviewed the concepts studied during the first week and prepared a short presentation for the engineering team. The presentation included cloud infrastructure, virtualization, and the main ideas I had learned from the documentation. I used the presentation to identify areas where my understanding was still incomplete. The feedback from the team helped me correct several misunderstandings.
Approbation de l'encadrant		*Signature et cachet de l'entreprise*

Bilan des activités de la semaine 1

Bilan du stagiaire
Description de la méthodologie de travail, des techniques et des outils utilisés …… …… …… …… …… …… ……
Principaux résultats obtenus …… …… …… …… ……

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Principales difficultés rencontrées …… …… …… …… …… ……
Commentaires de l'encadrant
Jugement du rendement du stagiaire, sa motivation et son initiative …… …… …… …… ……

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Activités journalières du stagiaire

Semaine 2

Journée	**Date** (jj/mm)	**Résumé des activités**
Lundi	06/07	I reviewed the project documentation and confirmed the final deliverables.
Mardi	07/07	I carried out a final check of the system architecture and validation results.
Mercredi	08/07	I prepared the final summary of the work completed and noted the remaining improvements.
Jeudi	09/07	I started studying cloud-native applications and Kubernetes. I reviewed the Kubernetes architecture, including the control plane, worker nodes, pods, and services. I also studied the role of Kubernetes in managing containerized applications. The objective was to understand Kubernetes before starting the practical deployment.
Vendredi	10/07	I continued researching Kubernetes and focused on its main objects and resources. I studied pods, deployments, services, namespaces, volumes, and scheduling. I also started learning about PersistentVolumes and PersistentVolumeClaims because storage would be an important part of the project. I documented the differences between temporary pod storage and persistent application storage.

Samedi	11/07	I started working on the project-management side of the internship. I discussed project organization and methodology with the project-management team. We considered the project scope, dependencies, available time, and expected deliverables. I decided to organize the work using a structured sequence from requirements to design, deployment, testing, and documentation.
Approbation de l'encadrant		*Signature et cachet de l'entreprise* Huawei Technologies Tunis RNE 0745097R Business Center 6ème étage les Berges du Lac 1053 La Marsa Service des Ressources Humaines

Bilan des activités de la semaine 2

Bilan du stagiaire
Description de la méthodologie de travail, des techniques et des outils utilisés …… …… …… …… …… …… …… ……
Principaux résultats obtenus …… …… …… …… …… ……
Principales difficultés rencontrées …… …… …… …… …… …… ……

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Commentaires de l'encadrant
Jugement du rendement du stagiaire, sa motivation et son initiative ....
Date : ..... / *Signature :* Huawei Technologies Tunis RNE 0745097P Business Center 6ème étage les Berges du Lac 1053 La Marsa Service des Ressources Humaines

Activités journalières du stagiaire

Semaine 3

Journée	**Date** (jj/mm)	**Résumé des activités**
Lundi	13/07	I started the requirements phase of the project. I identified the functional requirements related to the application and infrastructure. I also documented non-functional requirements such as high availability, security, automation, monitoring, and testing. These requirements were used as a reference for the architecture design.
Mardi	14/07	I continued working on the requirements documentation. I reviewed the relationship between the application requirements and the infrastructure required to support them. I prepared initial software and business requirements documents. I also discussed unclear points with the engineering team before moving to the
Mercredi	15/07	I continued working on the requirements documentation. I reviewed the relationship between the application requirements and the infrastructure required to support them. I prepared initial software and business requirements documents. I also discussed unclear points with the engineering team before moving to the
Jeudi	16/07	I started preparing the Software Design Document. I described the main physical and logical components of the proposed infrastructure. I also began documenting the network architecture and the relationship between the VMware and Kubernetes layers. This document became the main technical reference for the deployment work.
Vendredi	17/07	I continued the network design and studied data-center networking concepts. I researched spine-leaf architecture, layer-3 switching, routing, and ECMP. I also studied how physical network designs can be represented using virtual network components.

		This helped me prepare the network topology that would later be reproduced in the laboratory
Samedi	18/07	I continued studying virtual and overlay networking. I focused on how virtual machines communicate through virtual switches and how logical networks are built on top of physical infrastructure. The main challenge was separating traditional operating-system networking concepts from hypervisor networking. I updated the Software Design Document with the conclusions from this research.
Approbation de l'encadrant		*Signature et cachet de l'entreprise*

Bilan des activités de la semaine 3

Bilan du stagiaire
Description de la méthodologie de travail, des techniques et des outils utilisés …… …… …… …… …… …… …… ……
Principaux résultats obtenus …… …… …… …… …… ……
Principales difficultés rencontrées …… …… …… …… ……

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Commentaires de l'encadrant
Jugement du rendement du stagiaire, sa motivation et son initiative …… …… …… …… …… ……
Date : ……..
Signature : 

Activités journalières du stagiaire

Semaine 4

Journée	**Date** (jj/mm)	**Résumé des activités**
Lundi	20/07	I reviewed the architecture with the engineering team and received feedback on the proposed design. I corrected several parts of the network and infrastructure design. I also reviewed the dependencies between the VMware, networking, and Kubernetes components. The objective was to avoid beginning the deployment before the architecture was sufficiently clear.
Mardi	21/07	I reviewed the architecture with the engineering team and received feedback on the proposed design. I corrected several parts of the network and infrastructure design. I also reviewed the dependencies between the VMware, networking, and Kubernetes components. The objective was to avoid beginning the deployment before the architecture was sufficiently clear.
Mercredi	22/07	I prepared the laboratory environment for the deployment phase. Because the infrastructure had to run on limited local hardware, I evaluated the available computers and their resources. I identified several constraints related to CPU, memory, networking, and virtualization. An additional older computer was obtained to make the laboratory environment more practical.
Jeudi	23/07	I investigated the networking limitations of the available hardware. ESXi required a compatible Ethernet adapter, while the available office environment did not provide the expected wired network access. I connected the relevant machines using Ethernet and investigated the compatibility of the network adapter. This was my first major hardware-related troubleshooting task during the deployment.
Vendredi	24/07	I started deploying vCenter Server on the ESXi environment. I created the required virtual machine and started the vCenter installation process.

		The deployment required several configuration changes because of compatibility and resource problems. I documented the problems and the different solutions that I tested.
Samedi	25/07	I continued troubleshooting the vCenter Server deployment. I investigated compatibility problems and reviewed the virtual machine configuration and resource consumption. A CPU temperature problem also appeared on the laboratory hardware, so I adjusted the configuration to reduce resource pressure. After several attempts, the vCenter environment became stable enough to continue.
Approbation de l'encadrant		*Signature et cachet de l'entreprise*

Bilan des activités de la semaine 4

Bilan du stagiaire
Description de la méthodologie de travail, des techniques et des outils utilisés …… …… …… …… …… …… …… ……
Principaux résultats obtenus …… …… …… …… …… ……
Principales difficultés rencontrées …… …… …… ……

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Commentaires de l'encadrant
Jugement du rendement du stagiaire, sa motivation et son initiative …… …… …… …… …… ……
Date : ……..   *Signature :* 

Activités journalières du stagiaire

Semaine 5

Journée	**Date** (jj/mm)	**Résumé des activités**
Lundi	27/07	I continued testing vCenter Server and reviewed its management interface. I created the initial datacenter object and prepared the environment for cluster configuration. I studied the difference between a datacenter object and a vSphere cluster. I also reviewed the role of HA and DRS in a VMware cluster.
Mardi	28/07	I created and reviewed the initial vSphere cluster configuration. I enabled vSphere HA and DRS as part of the high-availability design. I reviewed the cluster settings before applying them to the available hosts. This helped me understand how vSphere groups hosts and manages shared resources.
Mercredi	29/07	I investigated an issue related to adding an ESXi host to the cluster. The host could enter maintenance mode while it was also supporting the vCenter environment. I studied the dependency between the VMware management plane and host-level operations. This showed me why infrastructure changes must be planned carefully in a production environment.
Jeudi	30/07	I created additional ESXi virtual machines using VMware Workstation Pro. I configured the virtual machines to support nested ESXi virtualization. I also adjusted the Windows host settings because some virtualization features interfered with nested virtualization. This gave me more experience with running a virtualized VMware environment inside a workstation.
Vendredi	31/07	I continued configuring the nested ESXi hosts and their network connections. I used bridged networking so that the virtual ESXi systems could communicate with the laboratory LAN. I tested the connectivity between the virtual hosts and the existing infrastructure.

		I also documented the virtualization settings required for the environment.
Samedi	01/08	I started deploying the virtual layer-3 network infrastructure. I used an Arista vEOS virtual machine to reproduce a data-center switch in the laboratory. I studied the deployment process and configured the basic interfaces. I also reviewed the ZTP feature and disabled it because automatic provisioning was not required for the laboratory.
Approbation de l'encadrant		*Signature et cachet de l'entreprise*

Bilan des activités de la semaine 5

Bilan du stagiaire
Description de la méthodologie de travail, des techniques et des outils utilisés …… …… …… …… …… …… …… ……
Principaux résultats obtenus …… …… …… …… …… ……
Principales difficultés rencontrées …… …… …… …… ……

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Commentaires de l'encadrant
Jugement du rendement du stagiaire, sa motivation et son initiative …… …… …… …… …… ……
Date : ……..
Signature : 

Activités journalières du stagiaire

Semaine 6

Journée	**Date** (jj/mm)	**Résumé des activités**
Lundi	03/08	I continued configuring the virtual network topology. I created the required VLAN and trunk configurations between the virtual network devices. I tested each connection before adding the next one. This approach made it easier to isolate network problems during deployment.
Mardi	04/08	I connected the ESXi host to the virtual leaf switch and connected the leaf to the spine switch. I adjusted the required virtual networking security settings for the nested environment. I configured routed interfaces and loopback addresses and started testing communication. The basic underlay network was then operational.
Mercredi	05/08	I completed the initial network connectivity tests. I verified communication between the ESXi hosts, virtual switches, and routed network components. I reviewed VLAN tagging and the behavior of trunk interfaces. I also updated the architecture documentation to reflect the final laboratory topology.
Jeudi	06/08	I reviewed the complete VMware infrastructure before starting the Kubernetes phase. I checked the datacenter, cluster, ESXi hosts, virtual networking, and resource availability. I also compared the deployed environment with the Software Design Document. The main infrastructure was stable enough to begin Kubernetes integration.
Vendredi	07/08	I started preparing the Kubernetes environment on top of the VMware infrastructure. I reviewed the Kubernetes architecture and identified the nodes and services required for the project. I focused particularly on the relationship between Kubernetes workloads and the underlying VMware virtual machines. I also prepared the environment for the storage integration phase.

Samedi	08/08	I continued configuring the Kubernetes environment and checking node communication. I verified that the Kubernetes nodes could communicate correctly across the virtual network. I reviewed the Kubernetes service and pod networking needed for later tests. I also prepared the cluster for integration with vCenter.
Approbation de l'encadrant		*Signature et cachet de l'entreprise*

Bilan des activités de la semaine 6

Bilan du stagiaire
Description de la méthodologie de travail, des techniques et des outils utilisés …… …… …… …… …… …… …… ……
Principaux résultats obtenus …… …… …… …… …… ……
Principales difficultés rencontrées …… …… …… …… …… …… …… ……

Commentaires de l'encadrant
Jugement du rendement du stagiaire, sa motivation et son initiative …… …… …… …… …… ……

Date : ……	*Signature :* 

Activités journalières du stagiaire

Semaine 7

Journée	**Date** (jj/mm)	**Résumé des activités**
Lundi	10/08	I prepared the final technical presentation and reviewed the project with the engineering team. I focused on explaining the architecture and the reasons for integrating VMware and Kubernetes. I also presented the main troubleshooting cases encountered during the internship. The feedback was used to improve the organization and clarity of the final documentation.
Mardi	11/08	I completed the final documentation of the internship project. I reviewed the daily activities, architecture, configurations, testing results, problems, and solutions. I identified possible future improvements such as automation, monitoring, dedicated hardware, stronger security, and more advanced failure testing.
Mercredi	12/08	
Jeudi	13/08	…… …… …… …… ……
Vendredi	14/08	…… …… …… …… ……
Samedi	15/08	…… …… …… …… ……
Approbation de l'encadrant		*Signature et cachet de l'entreprise*

Bilan des activités de la semaine 7

Bilan du stagiaire	
Description de la méthodologie de travail, des techniques et des outils utilisés …… …… …… …… …… …… …… ……	
Principaux résultats obtenus …… …… …… …… …… ……	
Principales difficultés rencontrées …… …… …… …… …… …… …… ……	
Commentaires de l'encadrant	
Jugement du rendement du stagiaire, sa motivation et son initiative …… …… …… …… …… ……	
Date : ……	*Signature :*


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Activités journalières du stagiaire

Semaine 8

Journée	**Date** (jj/mm)	**Résumé des activités**
Lundi	17/08	…… …… …… …… ……
Mardi	18/08	…… …… …… …… ……
Mercredi	19/08	…… …… …… …… ……
Jeudi	20/08	…… …… …… …… ……
Vendredi	21/08	…… …… …… …… ……
Samedi	22/08	…… …… …… …… ……
Approbation de l'encadrant		*Signature et cachet de l'entreprise*

Bilan des activités de la semaine 8

Bilan du stagiaire	
Description de la méthodologie de travail, des techniques et des outils utilisés …… …… …… …… …… …… …… ……	
Principaux résultats obtenus …… …… …… …… …… ……	
Principales difficultés rencontrées …… …… …… …… …… …… …… ……	
Commentaires de l'encadrant	
Jugement du rendement du stagiaire, sa motivation et son initiative …… …… …… …… …… ……	
Date : ……	*Signature :* Huawei Technologies Tunis RNE 0745097R Business Center 8ème étage les Berges du Lac 1053 La Marsa Service des Ressources Humaines

Activités journalières du stagiaire

Semaine 9

Journée	**Date** (jj/mm)	**Résumé des activités**
Lundi	24/08	…… …… …… …… ……
Mardi	25/08	…… …… …… …… ……
Mercredi	26/08	…… …… …… …… ……
Jeudi	27/08	…… …… …… …… ……
Vendredi	28/08	…… …… …… …… ……
Samedi	29/08	…… …… …… …… ……
Approbation de l'encadrant		*Signature et cachet de l'entreprise*

Bilan des activités de la semaine 9

Bilan du stagiaire	
Description de la méthodologie de travail, des techniques et des outils utilisés …… …… …… …… …… …… …… ……	
Principaux résultats obtenus …… …… …… …… …… ……	
Principales difficultés rencontrées …… …… …… …… …… …… …… ……	
Commentaires de l'encadrant	
Jugement du rendement du stagiaire, sa motivation et son initiative …… …… …… …… …… ……	
Date : ……	*Signature :*

Activités journalières du stagiaire

Semaine 10

Journée	**Date** (jj/mm)	**Résumé des activités**
Lundi	31/08	…… …… …… …… ……
Mardi	01/09	…… …… …… …… ……
Mercredi	02/09	…… …… …… …… ……
Jeudi	03/09	…… …… …… …… ……
Vendredi	04/09	…… …… …… …… ……
Samedi	05/09	…… …… …… …… ……
Approbation de l'encadrant		*Signature et cachet de l'entreprise*

Bilan des activités de la semaine 10

Bilan du stagiaire	
Description de la méthodologie de travail, des techniques et des outils utilisés …… …… …… …… …… …… …… ……	
Principaux résultats obtenus …… …… …… …… ……	
Principales difficultés rencontrées …… …… …… …… …… ……	
Commentaires de l'encadrant	
Jugement du rendement du stagiaire, sa motivation et son initiative …… …… …… …… ……	
Date : ……..	*Signature :*

APPRECIATIONS DE L'ELEVE-INGENIEUR

L'élève ingénieur est appelé à nous retourner son évaluation du stage ainsi que son auto-évaluation en tant que stagiaire.

Un formulaire sera mis à votre disposition en ligne par la Direction des stages. Il faudra le remplir à l'issu de votre stage et avant votre soutenance, avec les rubriques des tableaux ci-après.

Evaluation du stage				
Qualité de l'accueil	☐ Excellent	☐ Bien	☐ Moyen	☐ Insuffisant
Adéquation du cadre de travail	☐ Excellent	☐ Bien	☐ Moyen	☐ Insuffisant
Disponibilité et encadrement	☐ Excellent	☐ Bien	☐ Moyen	☐ Insuffisant
Qualité de l'intégration dans l'entreprise	☐ Excellent	☐ Bien	☐ Moyen	☐ Insu
Définition et clarté de la mission	☐ Excellent	☐ Bien	☐ Moyen	☐ Insuffisant
Consistance du programme	☐ Excellent	☐ Bien	☐ Moyen	☐ Insuffisant
Adéquation aux objectifs du stage	☐ Excellent	☐ Bien	☐ Moyen	☐ Insuffisant
Adéquation aux compétences	☐ Excellent	☐ Bien	☐ Moyen	☐ Insuffisant
Degré de recommandation de l'organisme d'accueil	☐ Excellent	☐ Bien	☐ Moyen	☐ Insuffisant
Degré d'atteinte des objectifs du stage	☐ Excellent	☐ Bien	☐ Moyen	☐ Insuffisant
Pertinence de l'expérience humaine	☐ Excellent	☐ Bien	☐ Moyen	☐ Insuffisant
Pertinence de l'expérience technique	☐ Excellent	☐ Bien	☐ Moyen	☐ Insuffisant

Auto-évaluation du stagiaire				
Assiduité	☐ Excellent	☐ Bien	☐ Moyen	☐ Insuffisant
Discipline	☐ Excellent	☐ Bien	☐ Moyen	☐ Insuffisant
Autonomie	☐ Excellent	☐ Bien	☐ Moyen	☐ Insuffisant
Apprentissage	☐ Excellent	☐ Bien	☐ Moyen	☐ Insuffisant
Méthodologie	☐ Excellent	☐ Bien	☐ Moyen	☐ Insuffisant
Résultats obtenus	☐ Excellent	☐ Bien	☐ Moyen	☐ Insuffisant

Remarques particulières et suggestions d'améliorations

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Date : ……..	*Signature du stagiaire :*

Informations importantes

(à lire attentivement)

1. Ce journal de stage doit être soigneusement complété par le stagiaire et présenté à la fin de chaque semaine à l'encadrant du stage pour compléter ses commentaires et sa signature.
2. Le stagiaire est appelé à compléter la description des résumés des activités pour toute journée de stage. Toute journée non complétée sera considérée comme absence sauf indication de jour férié ou d'autorisation d'absence signée par le responsable du stage.
3. Indépendamment de l'évaluation du jury de soutenance des activités de stage, ce dernier sera considéré non valide dans les cas suivants :
   - *Remise d'un journal de stage mal complété par le stagiaire*
   - *Remise d'un journal de stage ne comportant pas les commentaires et les signatures de l'encadrant du stage*
   - *La non remise du journal de stage à la Direction des stages à SUP'COM.*
4. Pour toute information complémentaire concernant les stages techniques avant, pendant ou après le déroulement des stages, les élèves ingénieurs sont invités à contacter :

La Direction des Stages
SUP'COM, Cité technologique des Communications
2083 El Ghazala, Ariana, Tunisie

E-mail : direction.stages@supcom.tn